Yiyao (Jim) Wan

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Industry Experiences

Software Engineer @ Meta (New York, NY) 02/2024 - present

- Built infrastructure to ensure numerics correctness of Meta's Ads model on Meta's custom AI accelerator (MTIA), using Python and PyTorch-based tooling to detect numeric errors from large scale to small reproduction across kernels, graph optimizations, and runtime.
- Built a simulation framework leveraging Torch Dispatcher and a custom AOT Autograd backend to emulate kernel execution on next-gen chip environments and detect numeric errors before fabrication, reducing post-silicon redesign cost by $25M per tape-out.
- Designed a data-driven test generator using Torch Dispatcher to capture real production model shapes and dtypes and auto-generate kernel-level unit tests against emulated environments, achieving nearly 100% kernel coverage across all production models.
- Developed kernel-level numerics interception and debugging tooling using Torch Dispatcher, a custom AOT Autograd backend, and PyTorch FX to capture per-op inputs/outputs, detect non-determinism across repeated executions, and reduce large failing training graphs (~thousands of ops) to minimal reproducible subgraphs, reducing Ads model onboarding debugging time from weeks to hours.
- Built LLM-powered agents using Claude API to automate accuracy debugging workflows, including an agent that orchestrates graph reduction tooling to auto-minimize failing training graphs and a training debugging agent that triages production numeric failures end-to-end, reducing manual debugging effort by X hours per incident.
- Architected an end-to-end accuracy regression CI pipeline with automated data refresh, integrating kernel-level and graph-level validation checks for NaN, numerical divergence, and non-determinism, achieving sub-hour turnaround with zero false negatives on numerical regressions.

Software Engineer @ Google (Seattle, WA) 10/2023 – 02/2024

- Worked at Google Flume Team usability team to enhance distributed workload experience, resolving 5 customer complaints and implementing user-suggested improvements using React.

Software Engineer @ Bloomberg (New York, NY) 04/2023 – 10/2023

- Worked at Query Language Infra Team to deliver Bloomberg Query Language (BQL) reliably and created a multi-service regression testing dashboard with TypeScript, reducing the release cycle by 14 days.
- Worked at Cloud Native Notebook Training Team Optimized shared memory size for pod creations with Go on Kubeflow controller, reducing 40% ram usage for Bloomberg-GPT model training.

Part-Time Software Engineer Contractor @ Aven (Remote) 09/2022 – 03/2023

- Worked at the Fraud Detection Team to reduce number of fraud users and prevent money laundry.
- Created internal dashboards with **Retool** and **Typescript** to group suspicious accounts by credibility, prevented 338k dollar loss.
- Developed a new service for legal team to manage inactive accounts with **Retool** and **Postgres**, and implemented legal-document-auto-generation pipelines into account closure workflow with **Pug.js**.

Software Engineer Internship @ Meta (Menlo Park, CA) 06/2022 – 09/2022

- Worked at MTIA Team to accelerate Meta AI models' inference, targeting a 50% reduction in cost across AI models inference operations.
- Implemented a Python benchmarking library for MTIA chips on CPU by re-implementing PyTorch table batched embedding operators at the compiler level with using C++ and Python, verified numerical loss within 0.1 against a medium-size Meta recommended model.
- Implemented weight pruning tools with C++ and Python to enable weight sparsity research while identifying configuration issues.

Software Engineer Internship @ Google (Remote**)** 01/2022 – 04/2022

- Worked at Google Vacation Rental Group to secure data accuracy for millions of users by developing a monitoring pipeline in data ingestion service to detect price mismatch from upstream sources.
- Implemented an alert system with C++ to auto-generate tickets for data accuracy issues, reported 10 hazards during 2 weeks of monitoring, and enhanced internal testing by adding hazard reporting feature into Chrome extensions.

Software Engineer Internship @ Walmart E-Commerce (Remote) 06/2021 – 08/2021

- Worked at E-Commerce Cloud Team on product cost, utilizing a **Jenkins-liked platform for CI/CD pipeline** to deploy on Cloud Native-Platform.
- Enhanced dashboard query service with **React, Spring Boot Framework**, and **Kotlin** to support node-based queries and improve production data inquiries.
- Deployed and monitored **Kafka** re-ingestion service with an average of 1.17s response time for 1 million+ requests using **Kubernetes** and **Splunk**.

Education

University of California, San Diego – Computer Science (MS) 2021 - 2023
University of California, Santa Barbara – Computer Science (BS) 2018 - 2021

Awards

First Place in Microsoft Azure Responsible AI Hackathon (**EasyMed**) 04/2022 - 06/2022

Skills

Languages: Python, C++, TypeScript, Go

Frameworks: PyTorch, Kubernetes, NodeJ